TT-D-376D
August 22, 1975
SUPERSEDING
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May 10, 1957

FEDERAL SPECIFICATION

DIPENTENE, TECHNICAL
(FOR USE IN ORGANIC COATINGS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers the product generally known as "commercial dipentene" for use as a thinner and anti-skinning agent in organic coatings.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

Federal Standards:

Fed. Test Method Std. No. 141/GEN - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Test Method No. 141/4261 - Appearance of Transparent Liquids.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 56 - Flash Point by Tag Closed Tester.
D 156 - Saybolt Color of Petroleum Products (Saybolt Chromometer Method).
D 233 - Sampling and Testing Turpentine.
D 484 - Hydrocarbon Dry Cleaning Solvents.
D 801 - Sampling and Testing Dipentene.
D 804 - Naval Stores and Related Products.
D 891 - Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials.
D 1296 - Odor of Volatile Solvents and Diluents.
D 1476 - Heptane Miscibility of Lacquer Solvents.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials. The dipentene shall consist solely of one kind as specified (see 6.2 and 6.4) without admixture with any other kind of terpenic or nonterpenic material.

3.2 Quantitative requirements. The dipentene shall meet the requirements specified in table I when tested as specified in 4.4.3.

TABLE I. Quantitative requirements

Characteristics	Requirements	
	Minimum	Maximum
Specific gravity at 15.6°/15.6°C	0.845	0.865
Refractive index at 20°C	1.472	1.483
Flash point	43°C (109°F)	---
Aniline point	---	15°C (59°F)
Distillation at 760 mm pressure		
Initial boiling point	168°C (334°F)	---
5 percent (by volume)	173°C (343°F)	---
95 percent (by volume)	---	188°C (370°F)
Residue after polymerization (percent by volume)	---	3.0

3.2.1 Color. The color of the dipentene shall be not darker than Saybolt No. +16 when tested as specified in 4.4.3.

3.3 Qualitative requirements.

3.3.1 Appearance. When tested as specified in 4.4.3, the dipentene shall be free from turbidity, suspended matter, and sediment.

3.3.2 Odor. When tested as specified in 4.4.3, the dipentene shall have an odor which is pleasantly mild and aromatic, suggestive of the characteristic lemon-like odor of the related terpene limonene and shall have only a faint aromatic residual.

3.3.3 Copper corrosion. When tested as specified in 4.4.3, the dipentene shall produce no more than a slight tarnish on the copper strip.

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3.3.4 Doctor test. When tested as specified in 4.4.3, the dipentene shall produce no black precipitate and no discoloration and shall not noticeably mask the yellow color of the sulfur film.

3.3.5 Spot test. When tested as specified in 4.4.3, the dipentene shall show no evidence of stain or residual oiliness.

3.3.6 Water test. When tested as specified in 4.4.3, the dipentene shall remain clear and produce no turbidity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. For the purposes of sampling, a lot of the dipentene shall consist of a manufacturer's batch. A batch is defined as the end product of all raw materials mixed, blended, or processed in a single operation.

4.3 Inspection.

4.3.1 Inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) of 2.5 percent defective to verify compliance with this specification regarding fill, closure, and marking and other requirements not involving tests. Each sample filled container shall be examined for defects of construction of the container, and the closure for evidence of leakage and for unsatisfactory markings; each filled container shall also be gaged to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.2 Inspection of preparation for delivery. The requirements of section 5 shall be examined and tested in accordance with the quality assurance provisions of PPP-P-1892.

4.4 Testing of the end item.

4.4.1 Sampling for testing. For the purposes of sampling, the lot shall be expressed in units of gallons of dipentene. The sample unit for testing shall be one quart of dipentene randomly selected from containers in the lot. The dipentene shall be placed in separate clean, dry metal or glass containers, sealed, marked, and forwarded to the testing laboratories. The sample size shall be as follows:

Lot size (gallons)	Sample size (quarts)
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

Tests shall be performed on a sample unit basis. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample unit fails to meet any test requirement specified.

4.4.2 Standard test conditions. Unless otherwise specified, all test specimens shall be prepared and tested in a room having a temperature of 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 5 percent.

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4.4.3 Test methods. All tests shall be conducted in accordance with the methods specified in table II to determine compliance with the requirements of 3.2 and 3.3.

TABLE II. Test methods

Test	Methods		
	ASTM Standard	Fed. Test Method Std. No. 141	Paragraph reference
Specific gravity	D 891		4.4.3.1
Refractive index	D 801, Section 8		
Flash point	D 56		
Aniline point	D 801, Section 15		
Distillation	D 801, Section 9		
Color	D 156		
Residue after polymerization	D 233, Sections 15-19		
Appearance		4261	
Odor	D 1296		4.4.3.2
Copper corrosion	D 801, Section 12		
Doctor test	D 484, Section 4.2.6		
Spot test	D 801, Section 11		
Water test	D 1476		4.4.3.3

4.4.3.1 Specific gravity. The specific gravity of the dipentene shall be determined at 15.6°/15.6°C (60°/60°F).

4.4.3.2 Odor. The dipentene, prior to testing, shall be contained in completely full and tightly stoppered containers without exposure to light until tested.

4.4.3.3 Water test. The ratio of the volume of heptane to the volume of dipentene shall be 19 to 1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The dipentene shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging shall be A, B, or C, and the level of packing shall be A, B, or C, as specified (see 6.2). The dipentene shall be furnished in 1-quart and 1-gallon containers, and 5-, 30-, or 55-gallon drums.

5.2 Special marking. In addition to markings specified in 5.1, information shall appear on each unit container and shipping container as follows:

(a) Date of manufacture (by month and year, not by code).
(b) Flash point in degrees Centigrade and degrees Fahrenheit.
(c) Any special directions for storage.

5.3 Precautionary marking. In addition to markings specified in 5.1, each unit container shall bear a label printed in bold type and securely affixed to the unit container as follows:

CAUTION:
DANGER - HARMFUL OR FATAL IF SWALLOWED
Contains dipentene.
Avoid frequent or prolonged contact with skin.
Use in a well ventilated area.
If swallowed, do not induce vomiting.
Call physician immediately.
Keep out of the reach of children.
Keep away from heat or open flame.
FLAMMABLE VAPOR HARMFUL

DO NOT USE IN OFFICE WORK OR IN BUILDING MAINTENANCE

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6. NOTES

6.1 Intended use. The dipentene covered by this specification is a refined terpene hydrocarbon and is intended for use as an adjunct to the thinner or solvent for certain types of organic protective coating material. It serves to prevent skinning in the can and to bring about better dispersion of the pigment in the vehicle. Due to the lower evaporation rate than for most other thinners, the presence of dipentene tends to retard surface drying of a paint or varnish film by keeping the surface of the film "open" longer during the drying period. Under certain conditions, the odor of the dipentene may have a bearing on its suitability for a proposed use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Kind required (see 6.4).
(c) Size of container required (see 5.1).
(d) Levels of packaging and packing required (see 5.1).

6.3 Dipentene shall be purchased by volume, the unit being the U.S. gallon of 231 cubic inches at 21°C (70°F). The volume of dipentene may be determined by dividing the net weight of the contents by 7.10, this being the average weight of a gallon of dipentene at 70°F.

6.4 Dipentene is the chemical as well as the trade name for the optically inactive form of the terpene hydrocarbon limonene. Four closely related kinds of dipentene are manufactured commercially, having more or less similar physical and chemical properties, depending on the process by which the dipentene is recovered from the pine wood used in its manufacture. These are steam distilled, sulfate, destructively distilled, and chemically processed dipentene. Further description or definition of the kinds may be found in ASTM D 804.

Civil Agency Interest:

VA-DMS

Preparing Activity:

GSA-FSS

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